

Network Coherence – From the Manifestation of Reality to Cosmic Cycles

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ABSTRACT

This episode gathers and systematises the fundamental concepts related to **Network coherence**: its definition, its two forms (**local** and **absolute global**), how it generates **macroscopic reality**, and how, at its limit, it dissolves that reality in order to prepare a **new cycle**. The mathematics is integrated where necessary, to show that these ideas are not mere metaphors but can be expressed formally.

1. WHAT IS COHERENCE IN THE NETWORK

In VG, **coherence** is the state in which the vibrations of the nodes are synchronised and their **phases are aligned**. Mathematically, two nodes are coherent when the phase difference between them is **zero**:

$$\phi_i - \phi_j = 0$$

and when they are **topologically adjacent** in nodal space:

$$d_{\text{nodal}}(i, j) = 0$$

This definition is taken from VG13 and RTS24, where we showed that topological adjacency **does not depend on physical distance**.

To describe the degree of coherence of a region, we can use an **order parameter**:

$$\Phi = \frac{1}{N(N-1)} \sum_{i \neq j} (\hat{S}_i \cdot \hat{S}_j)$$

- $\Phi = 1 \rightarrow$ **absolute global coherence** (all spins are aligned).
- $0 < \Phi < 1 \rightarrow$ **local coherence** (spins are partially aligned, but variations exist).
- $\Phi \approx 0 \rightarrow$ **total decoherence**.

Coherence is the **binder** that makes the Network behave as an **organised whole**. Without it, there is no **stable metric**, no **3D space**, no **physical laws**.

2. THE TWO FORMS OF COHERENCE

Local coherence is what generates and maintains macroscopic **reality**. It is characterised by **synchronisation**, but not by **uniformity**. The nodes vibrate in phase, but still retain **local differences** in amplitude and coupling. The **Hamiltonian** that describes this state is:

$$H = \sum_i \left(\frac{p_i^2}{2} + \frac{1}{2} \omega_i^2 q_i^2 \right) + \sum_{i < j} K_{ij} q_i q_j + \sum_{i < j} \lambda_{ij} (\hat{S}_i \cdot \hat{S}_j) q_i q_j + \sum_{i < j} \gamma_{ij} (\hat{S}_i \times \hat{S}_j) \cdot q_i p_j$$

This form, introduced in VG6, describes the **vibrations of the nodes**, the **couplings between them**, and the **influence of spins**.

Absolute global coherence is the state in which **all nodes vibrate not only in phase, but with the same amplitude**, and the vibrational potential difference between any two nodes becomes **zero**. In this state, the order parameter $\Phi = 1$, and the Hamiltonian reduces to a **perfectly symmetric form**. **Space and time cease to exist** as distinct entities.

This state can be described by the **relational crystal potential**:

$$V_{\text{crystal}} = -K_{\text{max}} \cdot \sum_{i < j} (\hat{S}_i \cdot \hat{S}_j)$$

The **minimum** of this potential corresponds to the state of **maximum coherence**. Any **deviation from alignment** increases the potential energy and **prepares the transition**.

3. HOW REALITY EMERGES FROM THE NODAL SUBSTRATE

The **emergence** of macroscopic reality from the nodal substrate follows **four stages**:

1. **The state of pure potential** — the Network exists, but is not yet organised into space and time. Couplings exist, but do not generate a stable metric.
2. **The appearance of a coherent zone** — a group of nodes enters **phase resonance** ($\Delta\phi_{ij} = 0$), forming a region in which vibrations are **synchronised**. Here, **local space is born**.
3. **The stabilisation of geometry** — the coherent zone is maintained, couplings become **regular**, and **physical laws** appear as properties of this organisation.
4. **The appearance of observers** — in regions of sufficient complexity, *C_{self}* **configurations** can form: autonomous nodal configurations capable of **observing** the Network.

3D reality is therefore a **stable region** of the nodal substrate, in which Φ has an **intermediate value** — large enough to maintain the metric, but small enough to allow **local variations**.

4. WHAT HAPPENS WHEN COHERENCE BREAKS — THE DESTINIES OF BLACK HOLES

A black hole is a place where the **coherence of the Network reaches a limit**. There, the couplings reach such a density that the Network can no longer sustain ordinary geometry. This state is described by the **critical coupling density**:

$$\rho_c = \rho_0 \cdot (1 + \alpha \cdot \langle \hat{S} \rangle^2)$$

introduced in VG2 (Series 3). When the coupling density reaches ρ_c , the Network enters a state of **instability**.

In this region, coherence can follow **three paths** (see also VG3 – Network Buds and VG5 – Between Mass and Spin):

1. **Slow evaporation** — the black hole gradually loses mass through **Hawking radiation**, and eventually leaves behind an **inert shell** (dark matter).
2. **Transformation into a white hole** — the coupling arrow $\mathcal{A}_{ij}$ **reverses**, and **absorption becomes emission**. After exhaustion, a shell remains.
3. **Birth of a Network bud** — if ρ reaches an **extreme critical threshold**, the spins unfold and trigger a **local unpacking** (mini-Big Bang), giving rise to a **daughter Network**.

These three destinies are **possible responses** of the Network to the same question: what to do with a region in which coherence has reached its limit?

5. ABSOLUTE GLOBAL COHERENCE — A MECHANISM AT TWO SCALES

What happens inside a black hole at the critical threshold is the **same thing** that happens at the end of a cosmic cycle, when the **entire Network enters contraction**.

In both cases, $\Phi \rightarrow 1$: **all nodes vibrate in phase, with the same amplitude**. The Hamiltonian reaches a state of **maximum symmetry**, and **space and time cease to exist**.

In the case of the **cosmic cycle**, this state prepares a **new Big Bang**. In the case of the **black hole**, it prepares a **mini-Big Bang**.

The **mechanism is identical**. The only difference is the **scale**.

6. CONCLUSION

VG8 introduces **nothing new**. It **gathers, systematises, and deepens** what we have already said in previous episodes, integrating mathematics where necessary.

We have shown that:

- **Coherence** is the **binder** that makes reality stable.
- It can be **local** (reality-generating) or **absolutely global** (reality-dissolving).
- **Black holes** are places where coherence reaches its limit and can follow **three destinies**.
- The **same mechanism** — absolute coherence and the unpacking of spins — **repeats at all scales**.

If this text is read, I hope the reader feels that it is not just a theory, but an **invitation** to see the Network as a **living organism** — one that breathes through coherence and renews itself through cycles.